2D

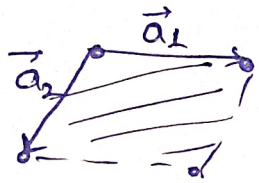
Crystal restriction theorem :- $1 + 2\cos\theta \in \mathbb{Z}$, where $\theta = \frac{2\pi}{n}$

$\Rightarrow n = 1, 2, 3, 4, 6$

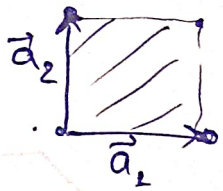
Allowed Point groups $= \{ \underbrace{C_1, C_2, C_3, C_4, C_6}_{\text{cyclic grp. (only rotation)}}, \underbrace{C_s, C_{2v}, C_{3v}, C_{4v}, C_{6v}}_{\text{rotations + reflections } (D_{2n} \text{ type})} \}$

~~(i) Square :-~~

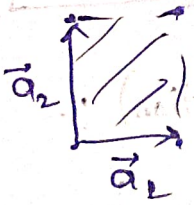
2D Bravais Lattice :-



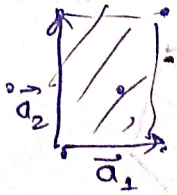
(oblique)



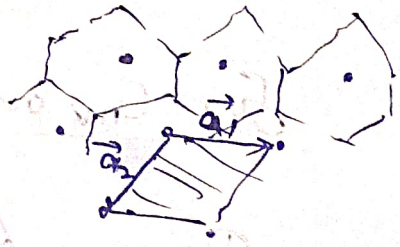
(Square) $(\vec{a}_1 = \vec{a}_2)$



(Primitive rectangular)

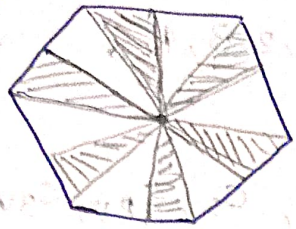


(centered rectangular)



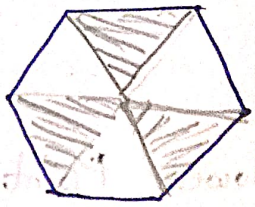
$(\vec{a}_1 \neq \vec{a}_2)$
(Hexagonal)

for Cyclic grps (C_n) :- (Rotation symm.)



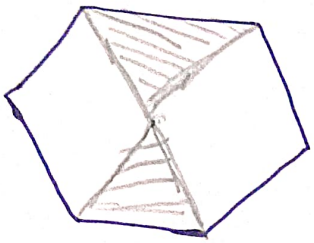
C_6

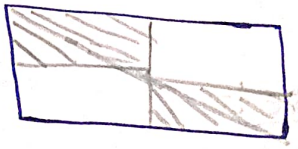
C_4

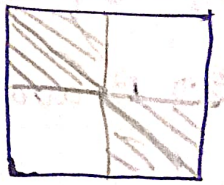


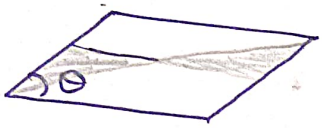
C_3

C_2

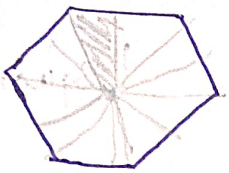


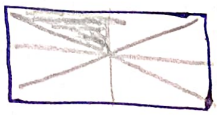


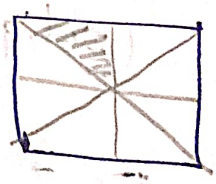




C_1.)

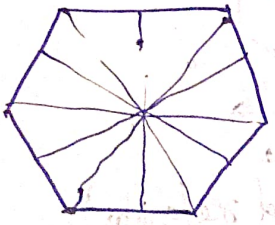






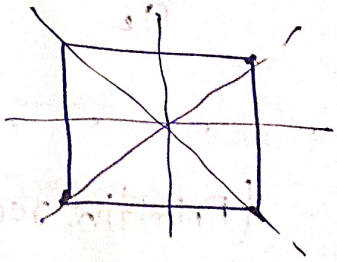
for C_{nv} or D_n groups :- (reflect^n + rotation) :-

D_6



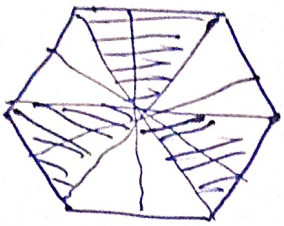
6 rot &
6 reflect^n.

D_4



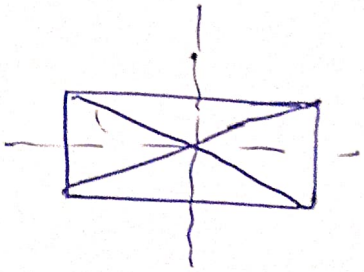
4 reflec^n. &
4 rotation.

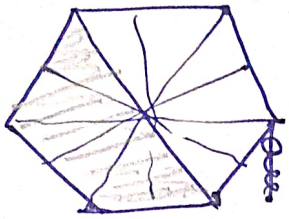
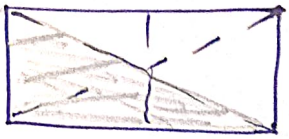
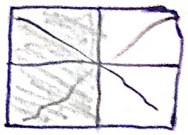
D_3 :



3 mirror &
3 rotation

D_2



2 mirror & 2 rotation

D_1